



Universidade do Minho
Escola de Engenharia



From Quantum Information to Quantum Networks

Architectures, Protocols, and the Road to the Quantum Internet

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About me

- 1992 - Founded my own business: CAD/CAE software development in partnership with Autodesk
- 1988 to 1993 – Electrical/Computer Engineering at Federal University of Uberlândia (UFU)
- 1989 to 1994 – Computer Science (UFU)
- 2002 to 2004 – Master's in Computing(UFU)
- 2005 to 2010 – participation in several software projects on different business domains (UOL, Globo.com, Algar Telecom, VIVO, TIM, Bunge, etc.)
- 2008 – Started at UFU as a Professor
- 2010 to 2013 – Ph.D. at the University of São Paulo
 - From Future Internet to Future Networks
 - Work at UFU in several research projects with industry and International collaboration
- 2023 – Started at University of Minho (Portugal)
 - European and international research collaboration
 - 5G/6G, AI-native & sustainable networks, with the research agenda expanding to include quantum networking
- Career snapshot:
 - 34 years of professional experience
 - 18 years in the industry;
 - 18 years in academia;
 - 14 years in internationally funded R&D projects in Europe and Brazil, supported by Horizon Europe, H2020, FP7, CAPES, CNPq, FAPEMIG, FAPESP and industry partners
 - **9 Ph.D. supervisions completed; 1 more expected within 3 months**
 - **22 M.SC. supervisions concluded**
- Research focus: Future Networks, IoT/loE, Network Softwarization (SDN/NFV), and Intelligent Networked Systems



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Part 1 —Why Quantum Networks?

Why talk about quantum networks now?



- The field is moving from architectural principles to applications, executable software and operational network architecture
- Quantum networking is becoming a systems and networking problem.

[1] [Kozłowski et al., Architectural Principles for a Quantum Internet, 2023](#)

[2] [Wang et al., Application Scenarios for the Quantum Internet, 2024](#)

[3] [Delle Donne et al., An operating system for executing applications on quantum network nodes, 2025](#)

[4] [Lopez et al., A Multiplane Architecture Proposal for the Quantum Internet, 2026](#)

Classical Internet vs Quantum Internet

Classical Internet

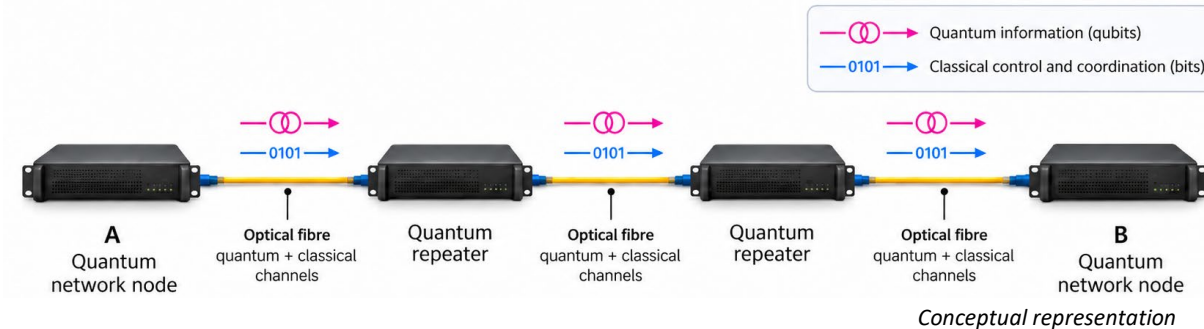
Transmits and processes classical bits



- Classical Internet
 - Information unit: **bit (0 or 1)**
 - Data can be **copied and regenerated**
 - Routers **inspect**, and **forward** information

Quantum Internet^[1,2,3]

Distributes quantum states and entanglement



- Quantum Internet
 - Information: **quantum states / qubits**
 - Unknown states **cannot be copied perfectly**
 - The network can **distribute entanglement**

Same need for connectivity — different information, different constraints

A Quantum Internet complements rather than replaces the classical Internet

- [1] [Wehner et al., Quantum internet: A vision for the road ahead, 2018](#)
 [2] [Cacciapuoti et al., Quantum Internet: Networking Challenges in Distributed Quantum Computing, 2020](#)
 [3] [Cameron et al., Entanglement swapping across a five-node relay in a multiplexed quantum-classical network, 2026](#)

What can quantum networks enable?

- Secure communication (QKD) is only one early application.
- The broader vision is to use shared quantum resources across a network.

QKD

distribute secret keys

Teleportation

transfer a qubit state

Distributed computing

connect remote QPUs

Distributed sensing

correlate remote sensors

Quantum cloud

networked quantum services

Future protocols

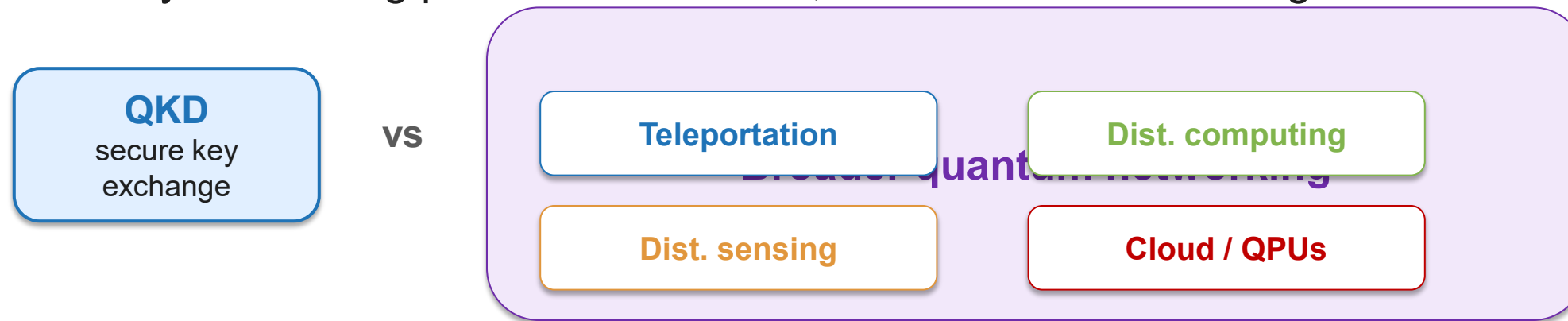
new applications built on entanglement

Quantum networking is about shared resources, not only secure links

Entanglement enables several classes of applications beyond QKD.

Quantum networking is not only QKD

- QKD is a valuable early use case, but it is not the whole story.
- Quantum networking also enables teleportation, distributed computing and sensing.
- The key networking problem is to create, store and deliver entanglement.

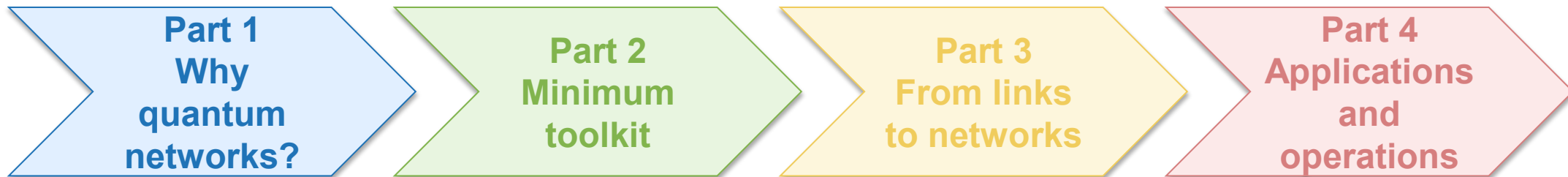


Quantum networking is not only QKD

The central network resource is entanglement, which supports many protocols.

Roadmap of the talk

- We first motivate the networking problem, then introduce only the quantum concepts we need.
- Next we move from links to multi-hop networks, and finally to applications and operations.



Roadmap: concepts first, networking next, applications last

The talk builds from the minimum quantum toolkit to the systems view.



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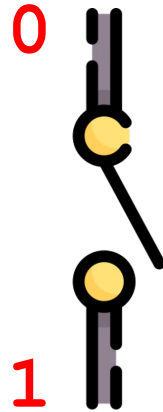


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Part 2 — Quantum Information: The Minimum Toolkit
Only the concepts we need to understand the networking problem.

From Bit to Qubit

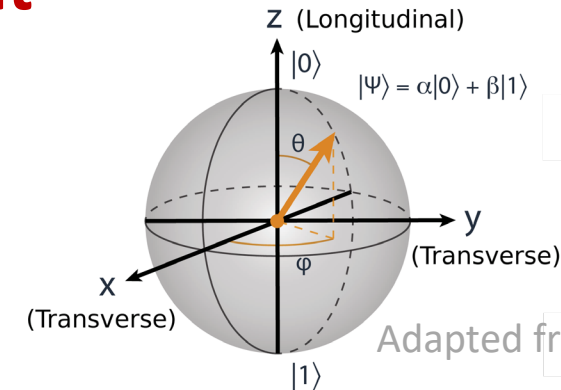
Classical bit



- State is **0 or 1**
- Can be **read without** fundamentally **changing** its logical value
- Can be **copied without** fundamental restriction

Qubit

(a) Bloch sphere



- Pure state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$
- **Measurement in the computational basis:** 0 with probability $|\alpha|^2$; 1 with probability $|\beta|^2$
- **Before measurement, the qubit may be in a superposition of the basis states**

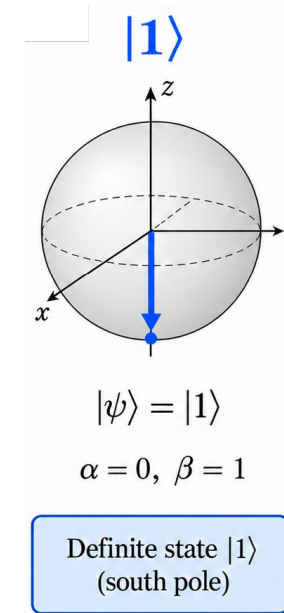
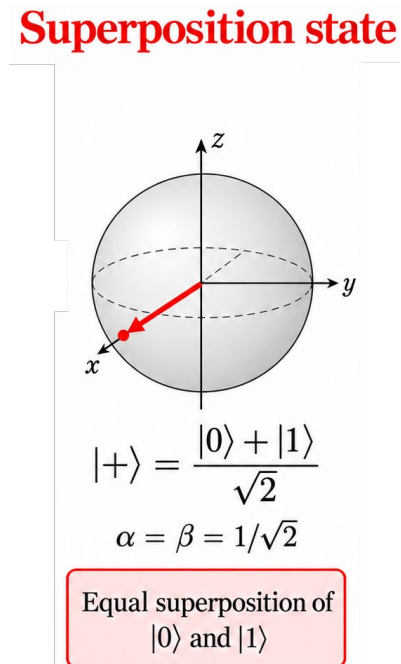
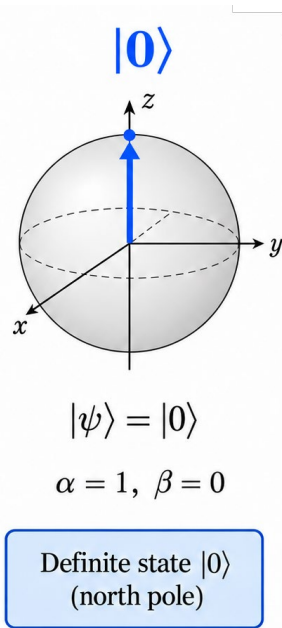
A qubit is not just a probabilistic classical bit

It is a quantum state that can exist in a superposition of $|0\rangle$ and $|1\rangle$.

[1] [Krantz et al., A quantum engineer's guide to superconducting qubits, 2019](#)

Superposition

- A qubit can be in a **superposition** of the basis states $|0\rangle$ and $|1\rangle$
- The state is described by **complex amplitudes** α and β , with $|\alpha|^2 + |\beta|^2 = 1$
- The amplitudes encode both **magnitude** and **relative phase**

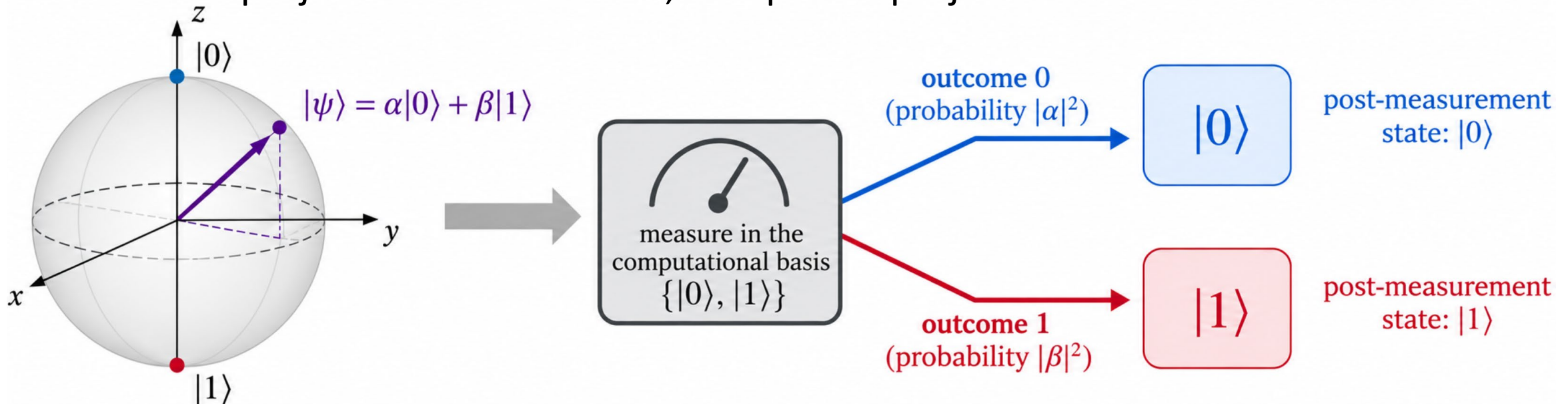


A superposition is a genuinely quantum state

It is not a classical uncertainty between 0 and 1, but a coherent combination of $|0\rangle$ and $|1\rangle$

Measurement

- Measurement produces a classical outcome
- In the computational basis, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$
- For an ideal projective measurement, the qubit is projected onto the observed basis state

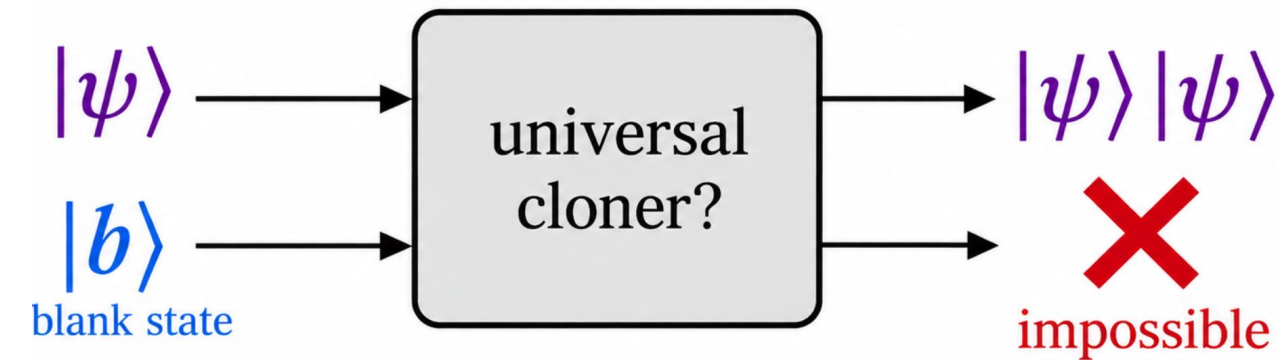


Measurement produces a classical outcome

It is not a passive read operation: measurement generally changes the quantum state

No-Cloning Theorem

- No universal operation can perfectly clone an arbitrary unknown quantum state
- This follows from the linearity of quantum mechanics



Assume cloning works for basis states^[1]

$$U|0\rangle|b\rangle \rightarrow |0\rangle|0\rangle$$

$$U|1\rangle|b\rangle \rightarrow |1\rangle|1\rangle$$

For $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

Linearity implies

$$U|+\rangle|b\rangle \rightarrow (|0\rangle|0\rangle + |1\rangle|1\rangle) / \sqrt{2}$$

But perfect cloning would require

$$|+\rangle|+\rangle = (|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle) / 2$$

Therefore,

$$(|0\rangle|0\rangle + |1\rangle|1\rangle) / \sqrt{2} \neq |+\rangle|+\rangle$$

So, a universal quantum cloner cannot exist.

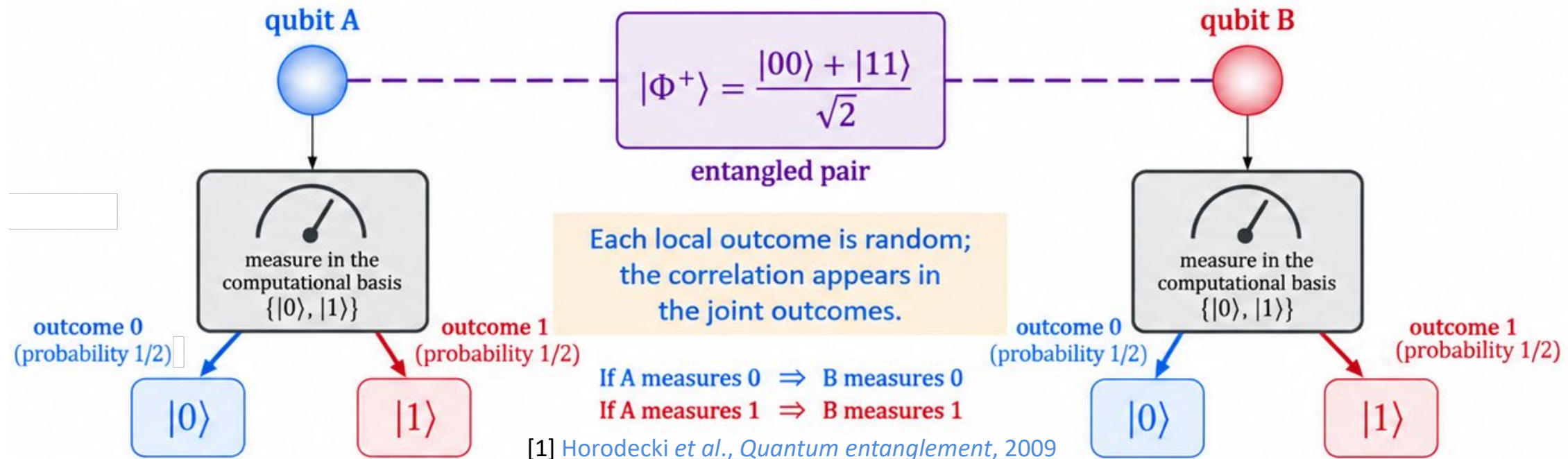
Arbitrary unknown quantum state cannot be copied perfectly

Quantum information cannot be replicated or regenerated as in classical networks.

[1] Wootters and Zurek, *The no-cloning theorem*, 2009

Entanglement

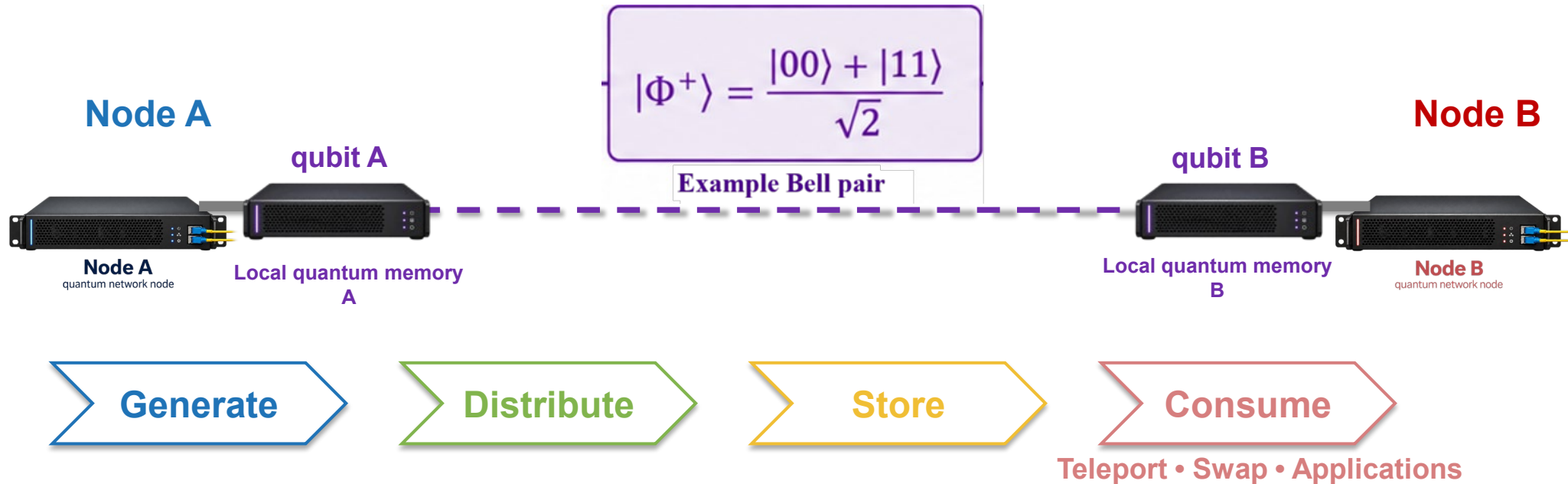
- Entanglement creates non-classical correlations between quantum systems
- The joint state cannot be separated into independent subsystem states
- For the Bell state $|\Phi^+\rangle$, measurements in the computational basis are locally random but perfectly correlated



Entanglement is a shared quantum-networking resource
It enables quantum protocols with no classical equivalent, including teleportation

Bell pair as a network resource

- A Bell pair is a shared entangled state distributed between two remote nodes
- Once established, it becomes a network resource that can be stored, managed, and consumed
- Entanglement-based applications consume Bell pairs as network resources

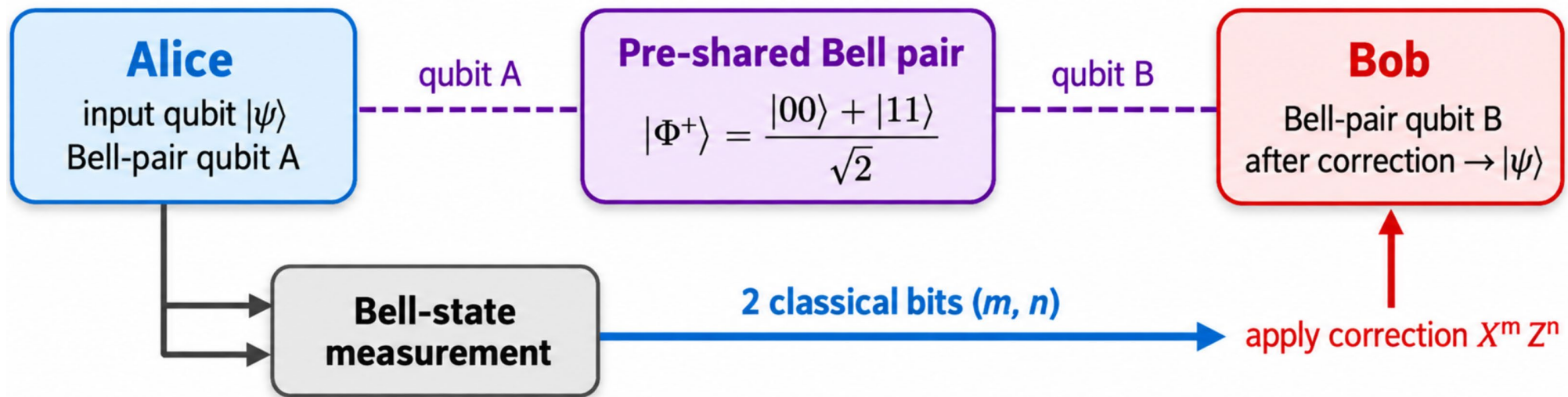


A Bell pair is a network resource, not just a physics object

The network generates, distributes, stores, and consumes Bell pairs on demand

Quantum teleportation: intuition

- Teleportation transfers an unknown qubit state using one shared Bell pair and two classical bits
- Alice measures the input qubit together with her half of the Bell pair
- Bob uses the two classical bits to recover the original state on his qubit



[1] [Hu et al., Progress in quantum teleportation, 2023](#)

Teleportation transfers a quantum state, not a physical qubit

It consumes one Bell pair and requires two classical bits

Why quantum networking is difficult

- Quantum networking is constrained by **loss**, **decoherence**, and the **no-cloning theorem**
- Many **entanglement-generation attempts** are **probabilistic** and **require coordination**
- Scalable long-distance networking therefore relies on **memories**, **repeaters**, and **classical control**

No-cloning theorem

unknown quantum states cannot be amplified or replicated

Loss

photons are lost in fibre and optical components

Decoherence

stored quantum states lose coherence while waiting

Probabilistic entanglement generation

successful entanglement attempts require classical confirmation (heralding)

Quantum networking must manage fragile, probabilistic resources

This drives the need for memories, repeaters, and classical coordination



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Part 3 — From Links to Networks

Building end-to-end quantum resources from fragile local links.

A quantum link is not yet a quantum network

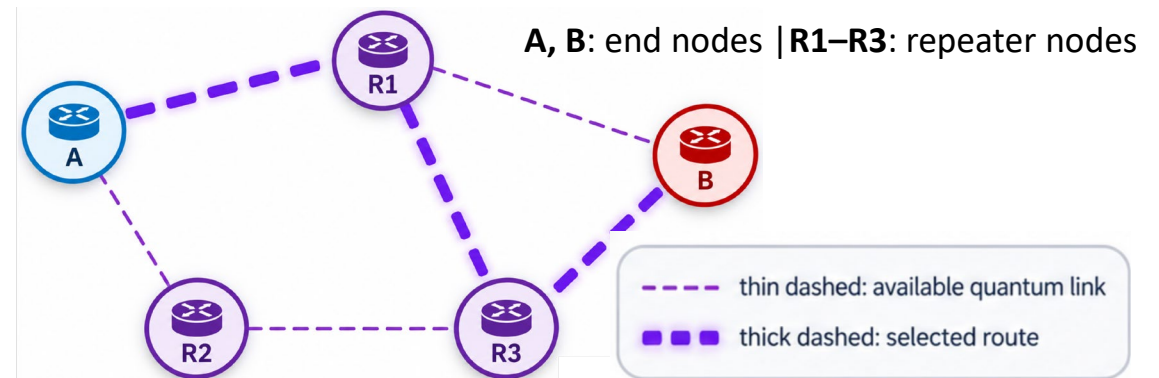
- A quantum link connects neighboring nodes for state transfer or entanglement generation
- A quantum network combines multiple links to create end-to-end quantum resources
- Networking adds routing, entanglement swapping, scheduling, and resource management

Point-to-point quantum link



Direct state transfer or entanglement generation

Multi-hop quantum network



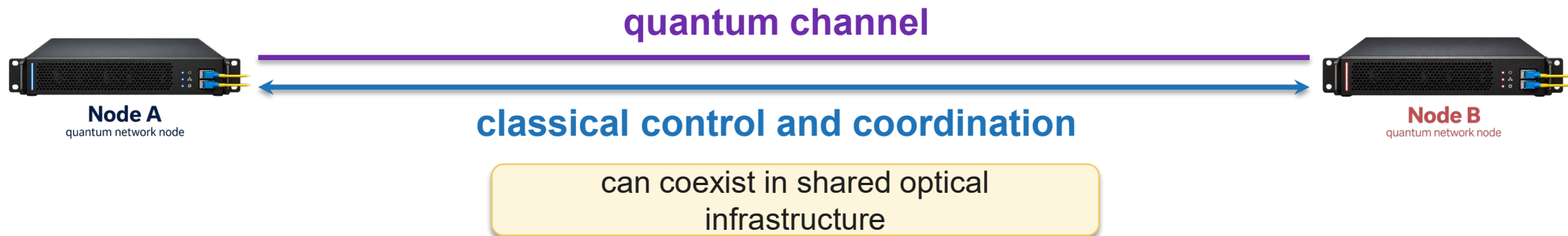
End-to-end entanglement via routing and swapping

A quantum link is a building block; a network creates end-to-end resources

The networking problem begins when multiple fragile links must be coordinated

Quantum channels and classical channels

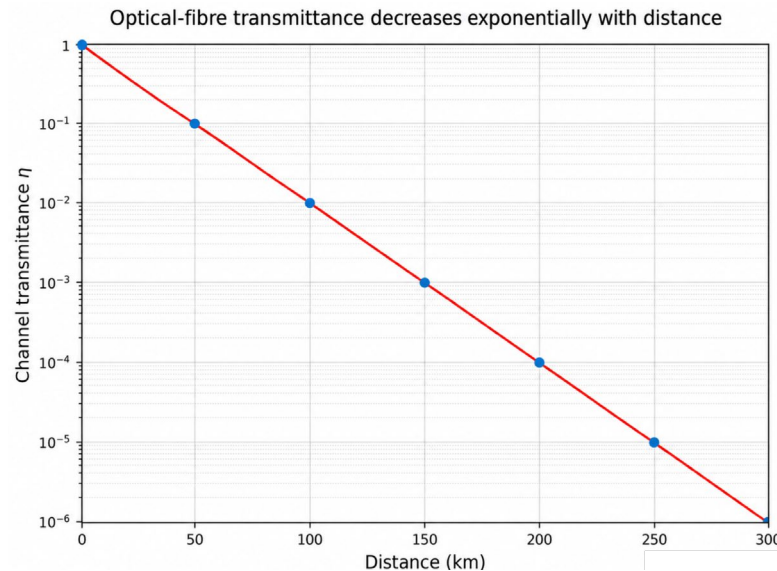
- Quantum channels carry qubits or quantum signals used to establish entanglement
- Classical channels carry success confirmation (heralding), synchronization, control, and correction messages
- Practical quantum networks require both quantum and classical communication



Quantum and classical communication are both essential
Practical quantum networks are inherently hybrid quantum–classical systems

Direct transmission does not scale

- Optical-fibre transmittance decreases exponentially with distance
- Unknown quantum states cannot be perfectly amplified or regenerated as in classical networks
- Long-distance communication therefore motivates quantum repeaters and entanglement-distribution architectures



Key implication

Loss accumulates, but unknown quantum states cannot be perfectly regenerated

Networking answer

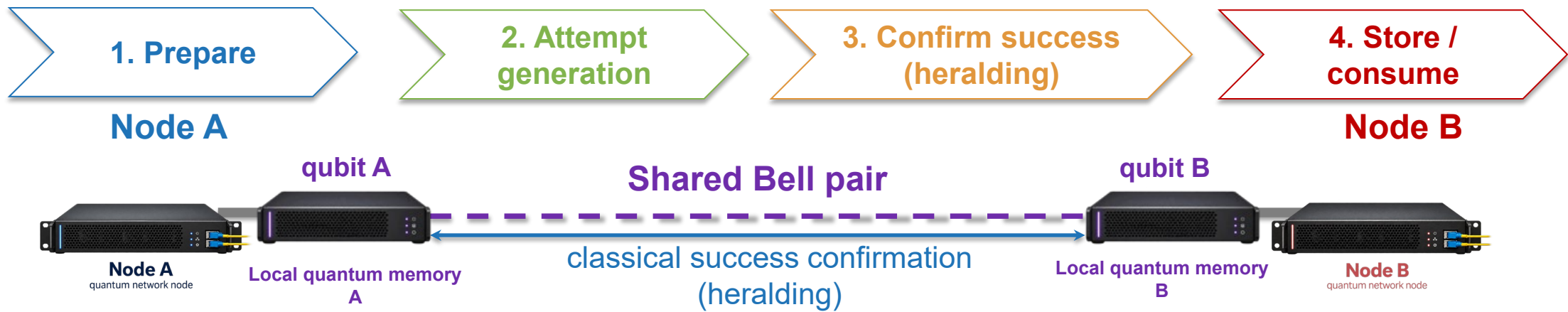
Establish entanglement over shorter elementary links, store the Bell pairs, and connect them using entanglement swapping

Direct quantum transmission does not scale to long distances

Loss and no-cloning turn distance into a networking problem

Entanglement distribution

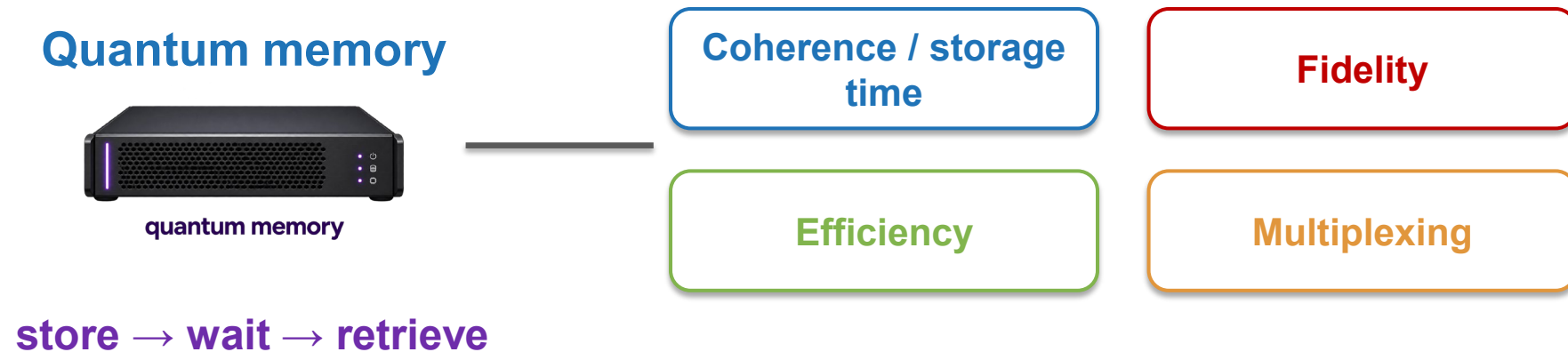
- Entanglement distribution establishes a Bell pair shared between remote quantum memories
- Generation may require repeated attempts; successful creation is confirmed by classical signaling
- The Bell pair can then be stored, consumed by an application, or used in entanglement swapping



Entanglement distribution is a fundamental quantum-network service
Many higher-level protocols begin by requesting a shared Bell pair

Quantum memories

- Quantum memories store qubits or entangled states while the network waits for other events
- They buffer successful entanglement while other links are established, enabling multi-hop protocols
- Important metrics include coherence time, efficiency, fidelity, and multiplexing

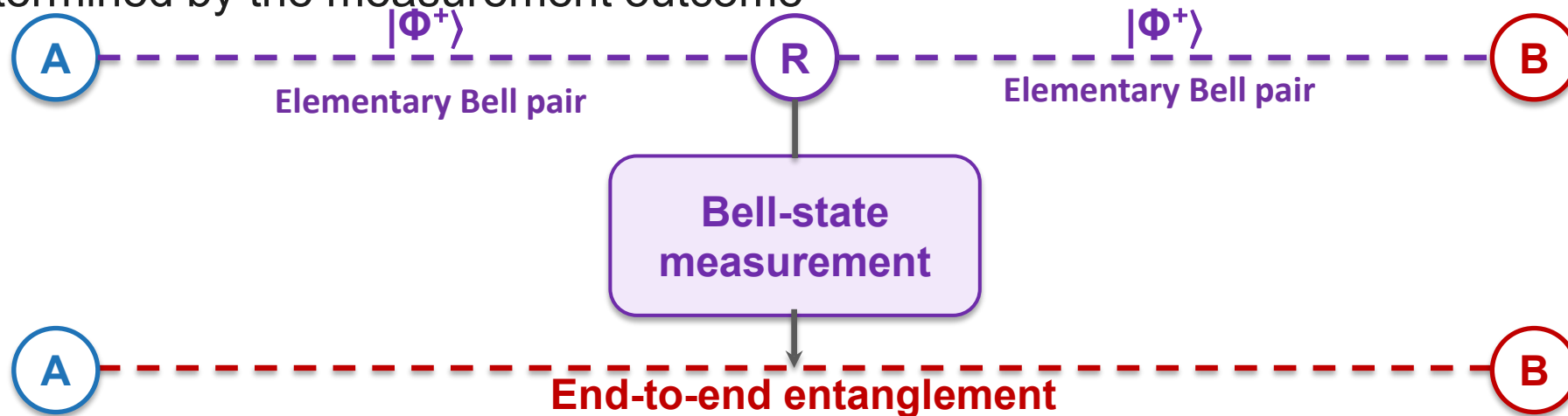


Quantum memories turn fragile link events into manageable network resources

Without storage, coordinating multi-hop entanglement would be much harder.

Entanglement swapping

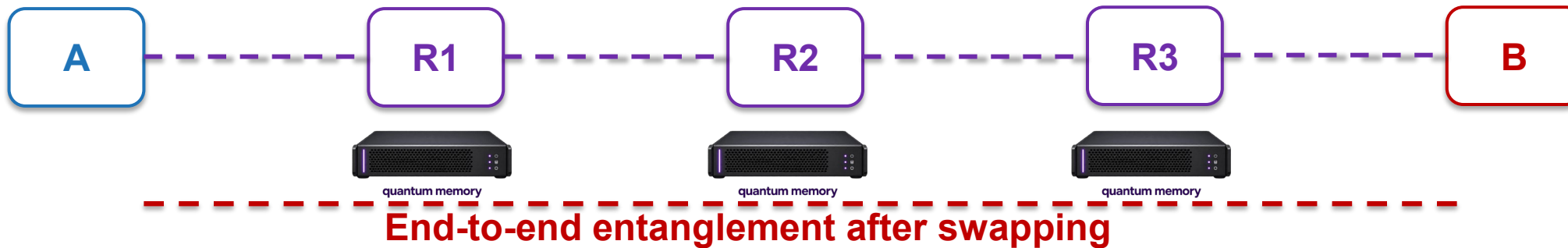
- Entanglement swapping connects two elementary Bell pairs into a longer-range Bell pair
- The middle repeater measures its two local qubits in a Bell basis
- After the Bell-state measurement, A and B share entanglement, up to a known correction determined by the measurement outcome



Entanglement swapping is the key mechanism for extending distance
It creates long-range connectivity out of short-range links.

Quantum repeaters

- Quantum repeaters extend range by combining **memories**, **entanglement generation**, and **entanglement swapping**
- They establish **elementary entangled links** and connect them through **entanglement swapping**
- Performance depends on **loss**, **memory lifetime**, and **the control strategy**



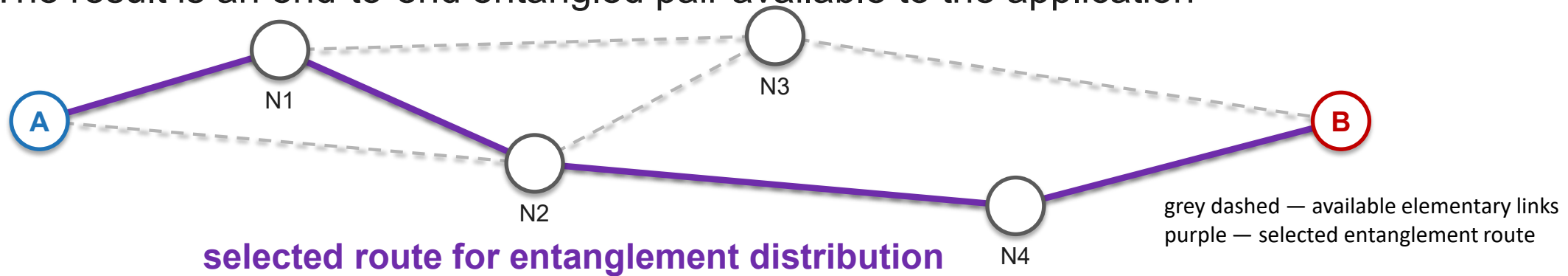
elementary links → stored pairs → swapping → longer-range entanglement

Quantum repeaters enable long-distance quantum networking

They turn many short, fragile links into a longer end-to-end resource

From links to end-to-end connectivity

- A multi-hop quantum path is built from many elementary links
- The network coordinates where to generate, store, and swap entanglement.
- The result is an end-to-end entangled pair available to the application



Multi-hop quantum connectivity is built from many local resources

This is why routing and coordination matter

Quantum-network architecture: first view

- A quantum network includes end nodes, repeaters, quantum links and classical control.
- Below the applications, the network must manage entanglement creation and delivery.
- This first architectural view separates hardware, network functions and applications.

Applications

Quantum network services

Control / routing / management

Hardware: nodes, memories, channels

Classical communication & support

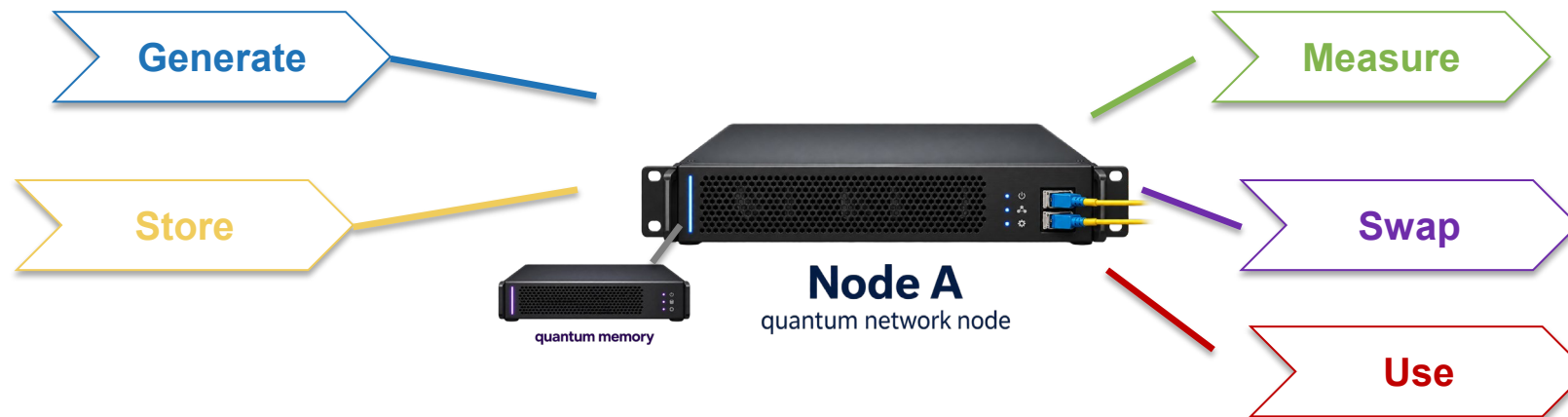
- signaling
- synchronization
- telemetry
- orchestration

A quantum network is a system of systems

Applications rely on a managed stack of hardware and network functions.

What does a quantum network node do?

- A node interfaces applications, local hardware and the network.
- It may generate, store, measure, swap and use entanglement.
- A local controller coordinates these operations with remote nodes.



A quantum-network node combines quantum hardware and classical control logic
Networking emerges from coordinated local operations across many nodes.

End nodes, repeaters, memories and control functions

- Different components play different roles in the architecture
- Scalable multi-hop quantum networks combine end nodes, repeater functions, memories, and classical control
- Thinking in functions helps separate the networking problem.

End nodes

run applications
and consume entanglement

Repeaters

extend distance
often via storage and
entanglement swapping

Memories

buffer fragile resources
while links are coordinated

Control functions

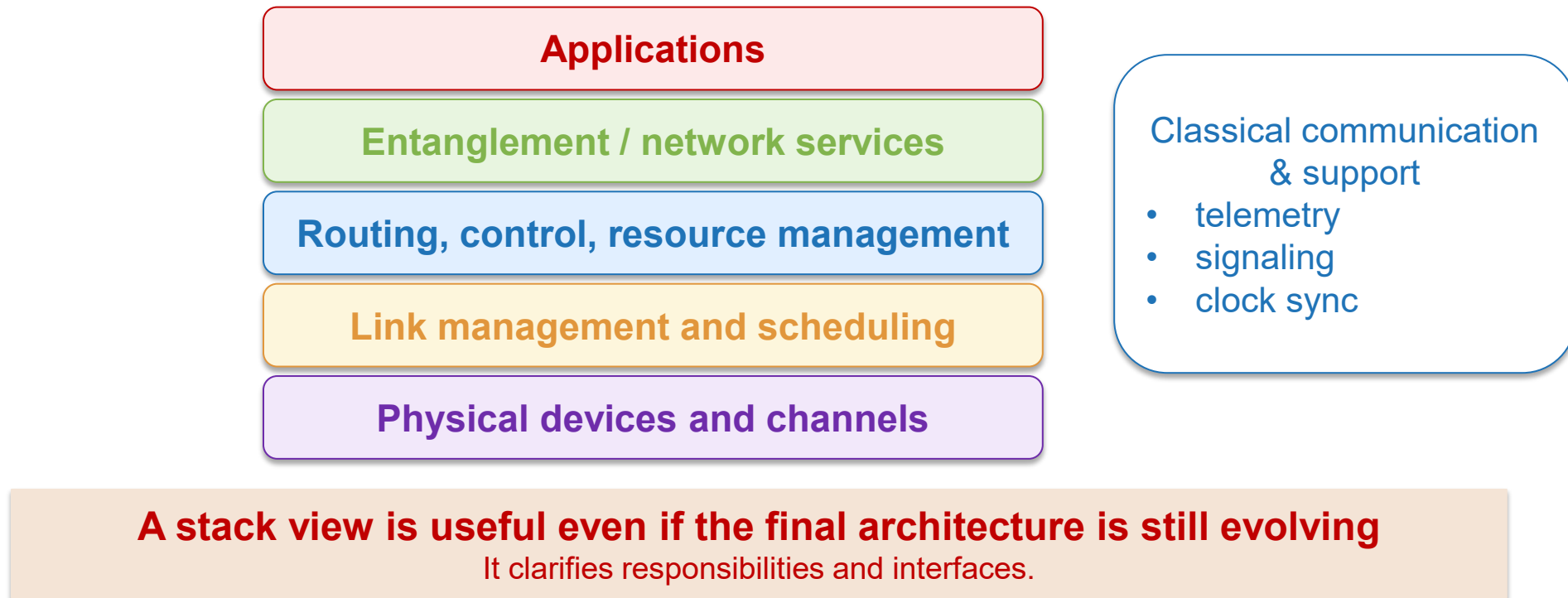
signal, schedule
and manage resources

Quantum networking needs both hardware roles and software/control roles

It is not only a physical-layer story.

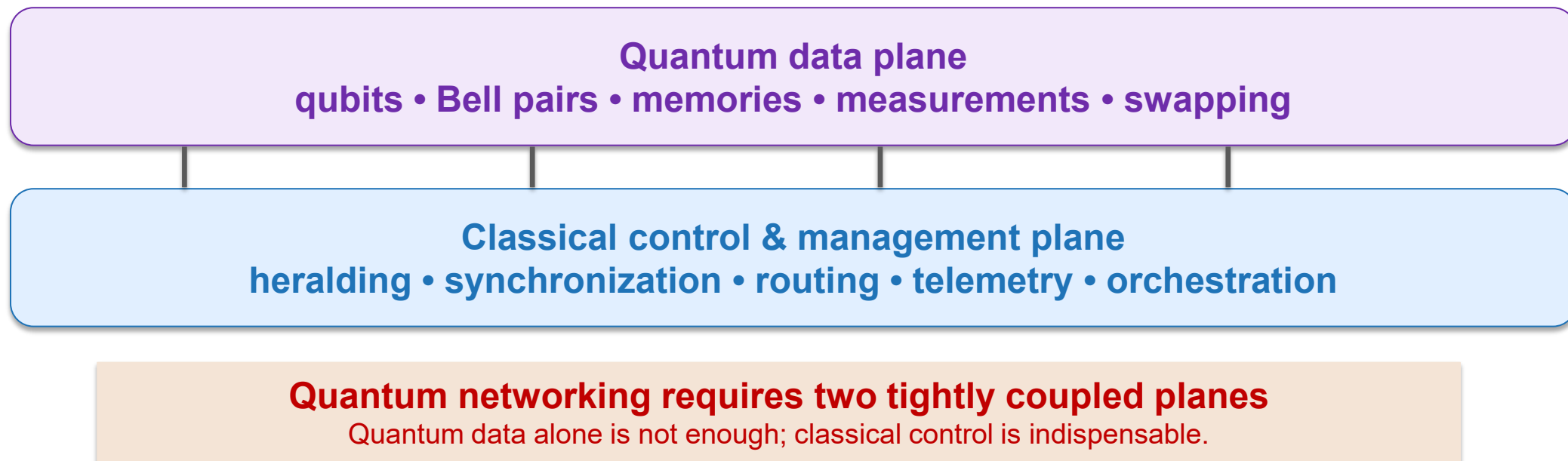
A conceptual quantum-network protocol stack

- A protocol stack helps separate physical mechanisms from services and applications.
- Lower layers manage link creation; upper layers expose network abstractions.
- The exact layering is still evolving, but the idea is useful.



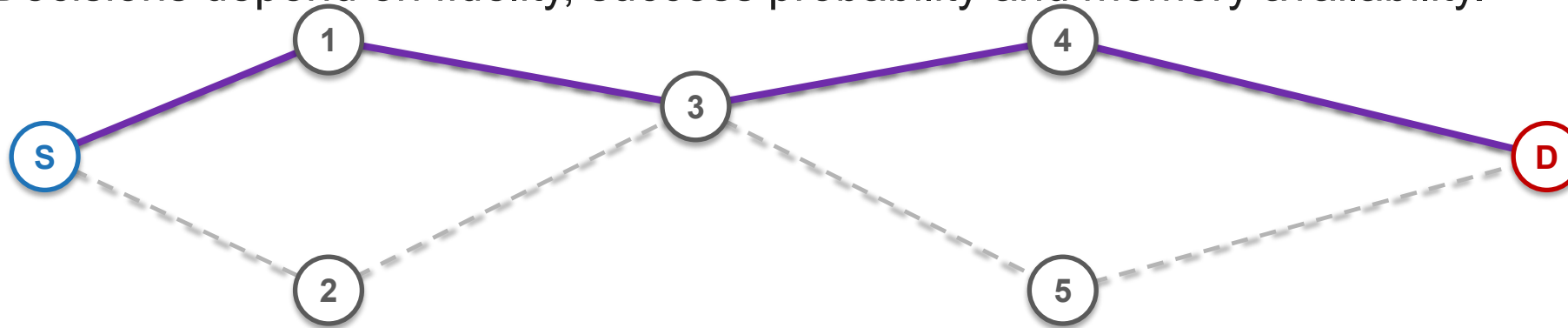
Quantum data plane and classical control plane

- The quantum data plane handles states, memories and measurements.
- The classical control and management plane coordinates signaling, synchronization, routing, telemetry, and orchestration.
- Both planes must cooperate tightly in real time.



Routing for entanglement distribution

- Routing for entanglement distribution differs fundamentally from packet routing
- The network selects a path and a schedule for creating end-to-end Bell pairs.
- Decisions depend on fidelity, success probability and memory availability.



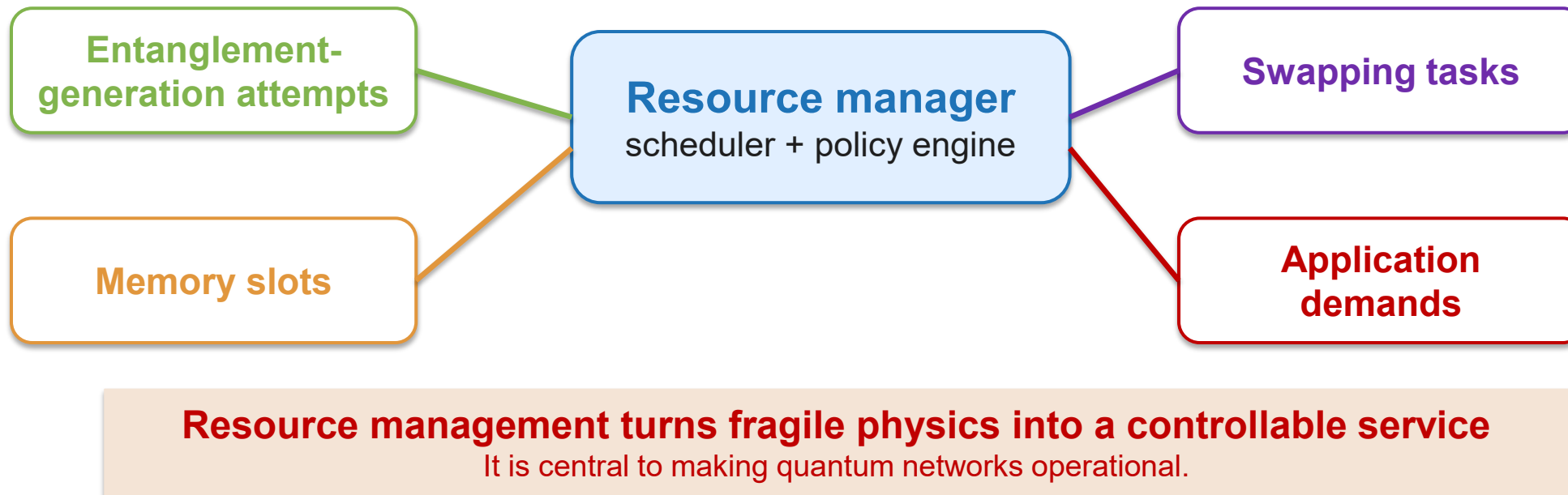
selected path maximizes end-to-end
utility

Routing determines how to build an end-to-end entangled resource, not simply where to forward packets

This is one of the conceptual shifts from classical networking.

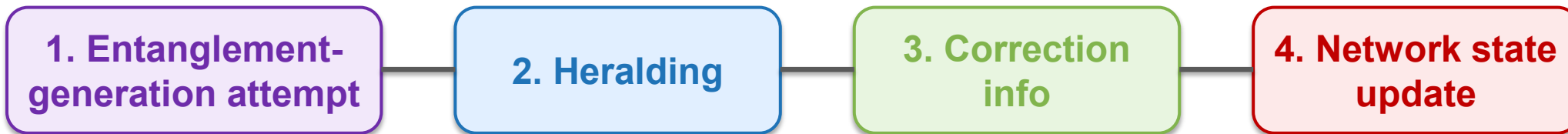
Resource management

- Resource management schedules entanglement-generation attempts, memory usage, and swapping operations
- It decides what to generate, reserve, consume, or discard
- This is a core systems problem in practical quantum networking.



Classical control is essential

- Classical control announces successful attempts and measurement outcomes.
- It also synchronizes operations, communicates correction information, and updates network resource state
- Without classical coordination, the quantum plane cannot operate reliably.

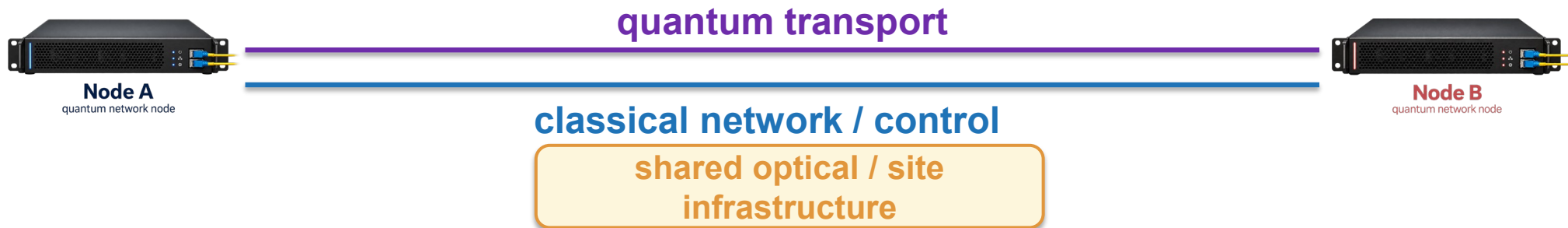


Classical control is not auxiliary; it is part of the protocol

Quantum networking is fundamentally quantum–classical co-design.

Hybrid quantum-classical infrastructures

- Near-term deployments are likely to integrate quantum devices with existing optical and IP infrastructures
- Fibre plant, sites, and management systems can be shared with classical networks
- Coexistence makes engineering, interoperability and operations crucial.



The likely reality is a hybrid quantum–classical infrastructure

Quantum networking will be deployed within existing network ecosystems.

Towards the Quantum Internet

- A plausible evolution is: isolated links → repeater-assisted networks → managed networks → programmable services
- Progress depends on co-evolution of hardware, protocols, software and operations.
- A long-term vision is a programmable Quantum Internet.



Toward the Quantum Internet means moving from devices to systems

Networking abstractions become more important as the field matures.



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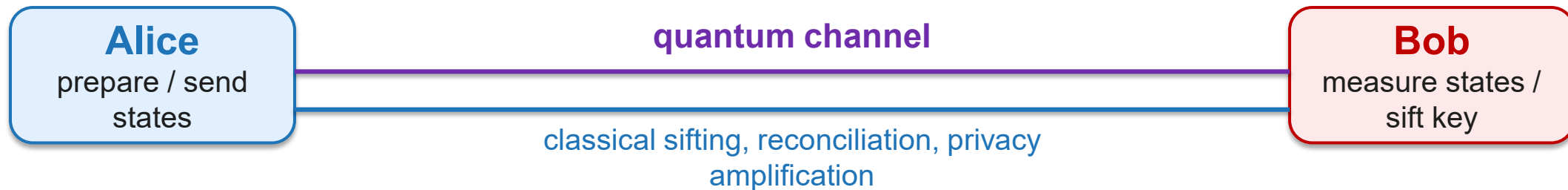
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Part 4 — Applications and Operations

From quantum-network resources to services, applications, and autonomous operation.

Quantum key distribution as an early application

- QKD establishes shared secret keys using quantum communication and authenticated classical post-processing
- It is an important early application, but it does not define the whole field of quantum networking.
- QKD can coexist with, and inform the development of, broader quantum-network infrastructures

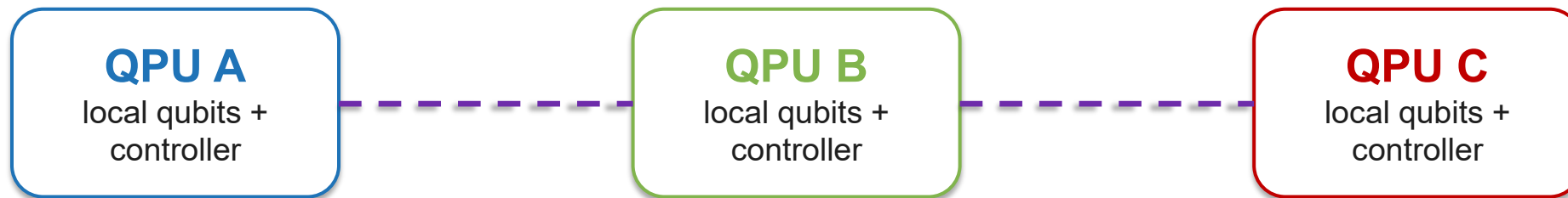


QKD is an early application of quantum communication, not the whole networking story

The broader agenda is about general shared quantum resources.

Beyond QKD: distributed quantum computing

- Remote Quantum Processing Units (QPUs) can share entanglement to execute distributed algorithms.
- Quantum networking can enable remote gates, state transfer, and modular quantum computation
- This is a major motivation for the Quantum Internet vision.



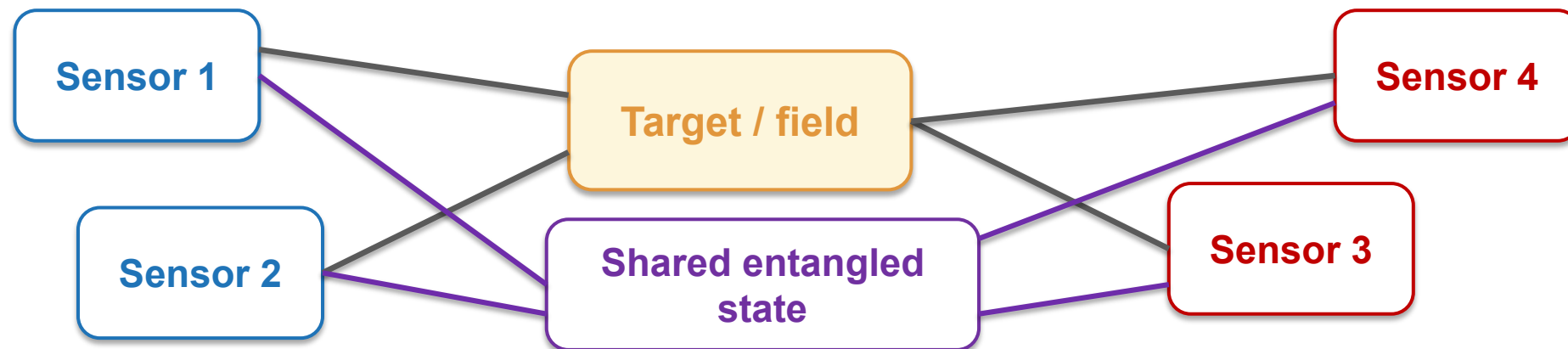
Shared entanglement can enable distributed quantum computation

Beyond QKD, quantum networking can connect remote processors

The network becomes part of the compute fabric.

Distributed quantum sensing

- Entanglement can improve the joint estimation performance of remote quantum sensors
- Applications include distributed metrology, clock comparison, and field sensing
- The network must deliver and manage shared quantum resources for sensing tasks.



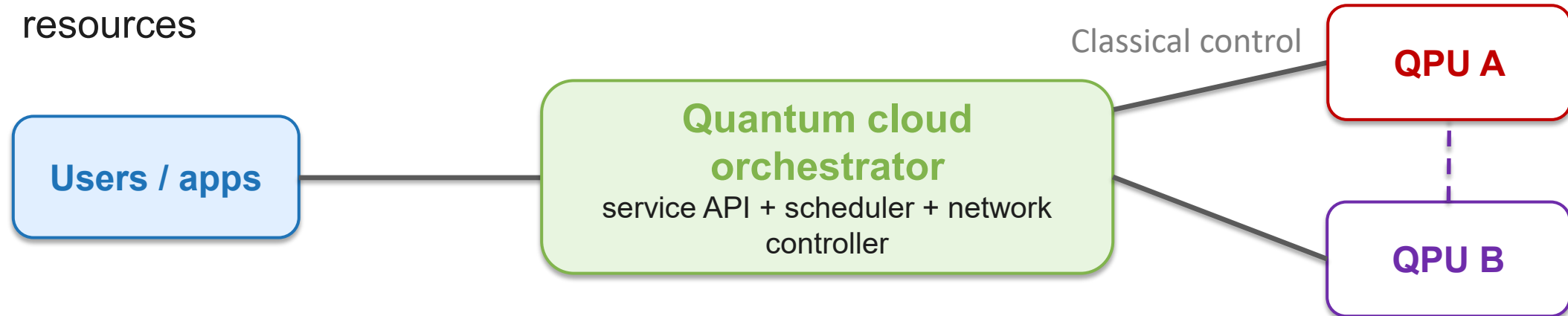
Shared entanglement can improve joint estimation

Distributed sensing is another class of network-enabled quantum application

The network distributes shared quantum resources among remote sensors

Quantum cloud and networked QPUs

- Future cloud services may connect multiple processors instead of exposing a single device.
- The network becomes part of the service fabric that links distributed QPUs and memories.
- Users request capabilities while orchestration abstracts the distributed hardware and network resources



Quantum cloud services may rely on networked hardware back ends

The network is part of the platform, not just a connection between sites.

Open challenges: distance, fidelity, memory, loss

- Major challenges span physics, systems and operations.
- They shape the feasibility and scalability of quantum-network services.
- These challenges are why the topic is now a real networking problem.

Distance

loss grows with length

Fidelity

noise limits usefulness

Memory

storage time and efficiency matter

Loss

no simple amplification

Scalability

many resources to coordinate

Interoperability

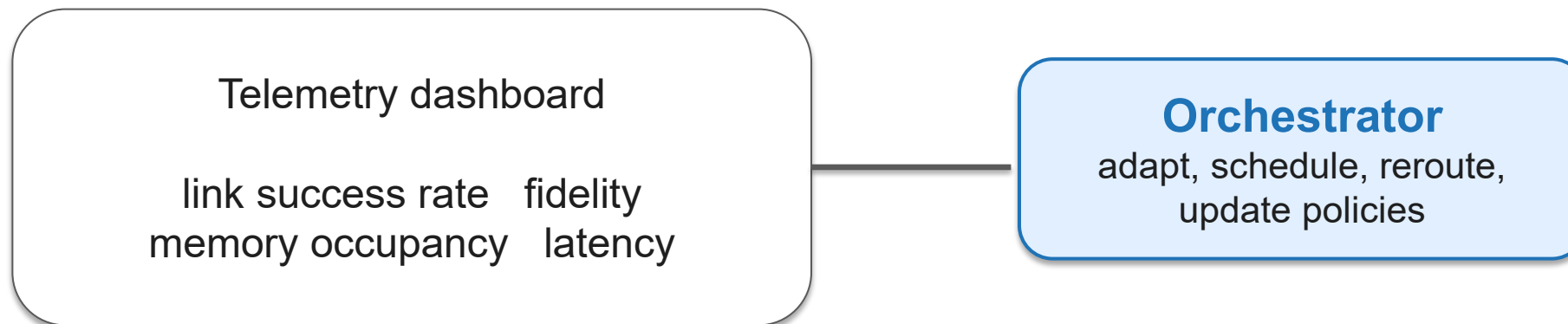
architectures and interfaces are evolving

Open challenges remain across the full stack

Quantum networking is hard because the physics and the systems issues interact.

Operating quantum networks: telemetry and orchestration

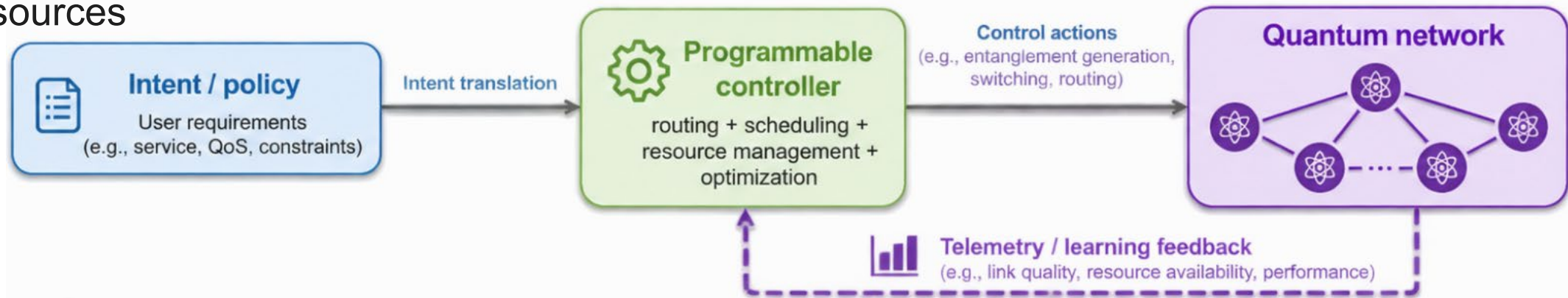
- Operating a quantum network requires visibility into both quantum and classical subsystems.
- Useful telemetry includes link success rate, fidelity, latency and memory occupancy.
- An orchestrator can adapt policies using this information.



Operations require measurement and control of the network itself
Telemetry and orchestration are becoming first-class concerns.

Towards programmable and increasingly autonomous quantum networks

- Programmability allows applications and operators to express network intent and policy
- Controllers translate intent into routing, entanglement generation, scheduling, and resource-allocation actions
- Telemetry and adaptive optimization can close the loop under uncertain and time-varying quantum resources



Programmability is a foundation for increasingly autonomous quantum-network operation

Intent, control actions, and telemetry form a closed feedback loop

Concluding Remarks

- Quantum networks distribute states and entanglement rather than replicated packets.
- Qubits, superposition, measurement, no-cloning and entanglement are the minimum conceptual toolkit.
- Memories, swapping, repeaters, routing and classical control turn fragile links into network services.
- The field is moving from principles to software, operations and applications.

Information model
qubits and entanglement

Core mechanisms
memories, swapping,
repeaters

Control
classical signaling is essential

Direction
toward programmable
networks

Takeaway: quantum networking is becoming a full systems and networking problem

That is why it is timely to discuss it now.



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Thank you!

Let's work together in the Quantum Network World?

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Department of Informatics
Engineering School
University of Minho



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Backup Slides

Entanglement swapping across a five-node relay in a multiplexed quantum-classical network¹

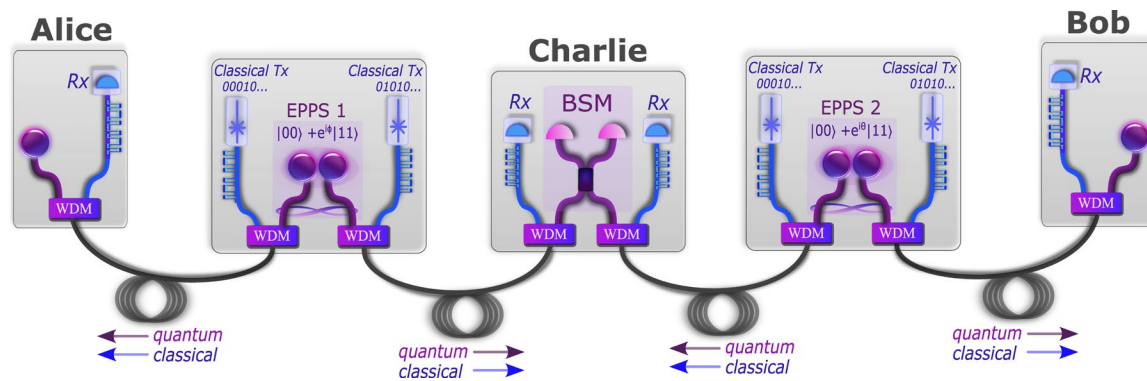


Fig1. Conceptual diagram of entanglement swapping across a five-node quantum relay

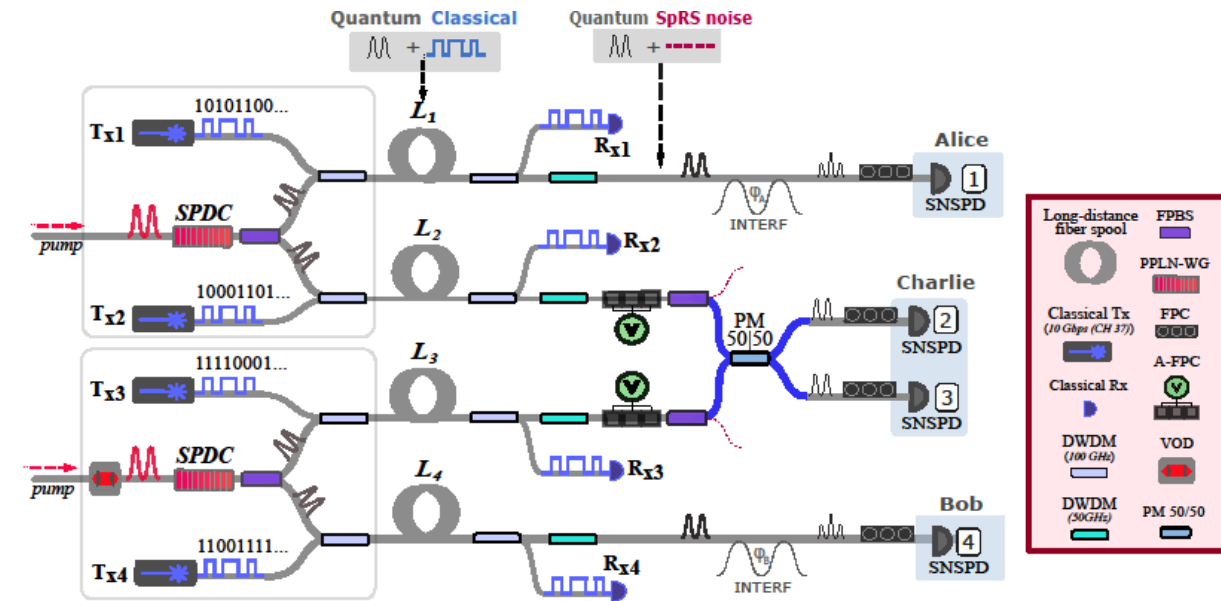


Fig2. Detailed experimental architecture for entanglement swapping in a quantum-classical coexistence relay

[1] [Cameron et al., Entanglement swapping across a five-node relay in a multiplexed quantum-classical network, 2026](#)